

MID TERM EXAMINATION

Subject: Linked Lists, Stacks, Queues

Paper: Mid Term Examination - Set A | Duration: 3 Hours | Max Marks: 37 | Date: 11-03-2026

Total Questions: 11 | Answer all questions.

Section 1: MCQ (6 Questions - 12 Marks)

Q1. [2 marks] (Medium)

Here are 10 unique exam questions for your Mid Term Examination in Data Structures (CS101):

Q2. [2 marks] (Easy)

Which of the following is NOT a characteristic of a singly linked list?

Options:

A) Each node stores a pointer to the next node. B) Elements are stored in contiguous memory locations. C) It is a dynamic data structure. D) Insertion and deletion can be efficient at the beginning.

Q4. [2 marks] (Easy)

In a queue data structure, which operation is used to add an element to the rear (back) of the queue?

Options:

A) Pop B) Enqueue C) Dequeue D) Peek

Q6. [2 marks] (Medium)

If a stack has a maximum capacity of 5 elements, and the operations push(A), push(B), pop(), push(C), push(D), pop() are performed in that order, what will be the top element of the stack at the end?

Options:

A) A B) B C) C D) D

Q8. [2 marks] (Medium)

In a singly linked list, which of the following operations generally has a time complexity of $O(N)$ in the worst-case scenario, where N is the number of nodes?

Options:

A) Inserting a node at the beginning. B) Deleting the last node. C) Accessing the first node. D) Inserting a node after a given node (pointer to given node is available).

Q10. [2 marks] (Hard)

A circular queue is implemented using an array of size 5. It uses front to point to the first element and rear to point to the last element. Initially, the queue is empty (front = -1, rear = -1). The following operations are performed: enqueue(10), enqueue(20), enqueue(30), dequeue(), enqueue(40), enqueue(50). What is the sequence of elements from front to rear in the queue at the end?

Options:

A) 20, 30, 40, 50 B) 30, 40, 50 C) 10, 20, 30, 40, 50 D) 50, 20, 30, 40

Section 2: Short Answer (5 Questions - 25 Marks)

Q3. [5 marks] (Medium)

Explain a real-world application where a stack data structure would be a suitable choice. Describe how the LIFO (Last-In, First-Out) principle is utilized in that application.

Q5. [5 marks] (Hard)

Describe a scenario where a doubly linked list would be preferred over a singly linked list, and justify your choice based on operational efficiency.

Q7. [5 marks] (Easy)

List the three primary operations typically associated with a queue data structure, and briefly describe what each operation does.

Q9. [5 marks] (Medium)

Describe how an array can be used to implement a stack. What are the potential limitations of an array-based stack implementation?

Q11. [5 marks] (Hard)

You are given two singly linked lists, L1 and L2, both of which are already sorted in non-decreasing order. Design an algorithm to merge them into a single sorted singly linked list. Explain the time and space complexity of your algorithm.

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